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PRE-CHARGING ARRANGEMENT, POWER CONVERTER SYSTEM, AND METHOD FOR PRE-CHARGING POLY-PHASE POWER CONVERTER

Abstract

A pre-charging arrangement (10) for a poly-phase power converter (20), optionally an H-bridge converter, is arranged for being supplied by a poly-phase AC power supply (40) via an input filter (30), wherein the poly-phase power converter (20) is arranged to be connected to and disconnected from the poly-phase AC power supply (40) by a first circuit breaker (35). The pre-charging arrangement (10) includes an AC power supply (12) connected via a transformer (14), having a primary side (14A) and a secondary side (14B), between a first common point (39) of the input filter (30) and a second common point (29) of the poly-phase power converter (20) for pre-charging DC link capacitors (25) of the poly-phase power converter (20), and a second circuit breaker (15) for connecting and disconnecting the pre-charging arrangement (10).

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims foreign priority benefits under 35 U.S.C. § 119 to German Patent Application No. 102024105831.5 filed on Feb. 29, 2024, the content of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates in general to power converters, such as frequency converters, inverters, or rectifiers. In particular, however, not exclusively, the present invention concerns pre-charging arrangements and methods for pre-charging poly-phase power converters, and utilization thereof in power converter systems. Such pre-charging arrangements and methods, and power converter systems may be used, for example, in battery energy storage systems, electric vehicle charging systems, power-to-X (especially hydrogen) electrolyser systems, and/or widely in frequency converter systems.

BACKGROUND

[0003] Without pre-charging, the current of power converter becomes at least momentarily very high when being connected to the grid or AC (alternating current) power supply in order to start normal operation. In existing pre-charge strategies for poly-phase converter topologies, especially for H-bridge rectifiers, a large number of pre-charge circuits, such as separate pre-charge circuits including a diode or switch rectifier for each H-bridge DC bus, are needed. This is costly and adds complexity to the systems. In addition to H-bridge converter topologies, the problem with very high current applies also to other power converter topologies. There is, thus, still a need to develop pre-charging arrangements and methods for power converters.

SUMMARY

[0004] An objective of the present invention is to provide a pre-charging arrangement, a power converter system, and a method for pre-charging a poly-phase power converter. Another objective of the present invention is that the pre-charging arrangement, the power converter system, and the method simplify pre-charging of the DC link capacitors of poly-phase power converters.

[0005] The objectives of the invention are reached by a pre-charging arrangement, a power converter system, and a method for pre-charging a poly-phase power converter as defined by the respective independent claims.

[0006] According to a first aspect, a pre-charging arrangement for a poly-phase power converter is provided. The poly-phase power converter is arranged for being supplied by a poly-phase AC power supply via an input filter, wherein the poly-phase power converter is arranged to be connected to and disconnected from the poly-phase AC power supply by a first circuit breaker.

[0007] The pre-charging arrangement comprises an AC power supply connected via a transformer, having a primary side and a secondary side, between a first common point of the input filter, preferably a star point formed by interconnected terminals of filter capacitors, and a second common point of the poly-phase power converter for pre-charging DC link capacitors of the poly-phase power converter. In addition, the pre-charging arrangement comprises a second circuit breaker or contactor for connecting and disconnecting the pre-charging arrangement.

[0008] The power converter may be, for example, a poly-phase H-bridge converter or a neutral point clamped (NPC) power converter.

[0009] In case of a poly-phase H-bridge converter, there may be one or several (cascaded) H-bridges per phase. The second common point in said cases is, preferable, the point where the (cascaded) H-bridge(s) from each phase (or phase legs) are connected to each other, thus forming a star point.

[0010] In case of an NPC power converter, the second common point is, preferable, the neutral point or the DC midpoint (a point between series connected DC link capacitors) of the converter. An NPC power converter preferably comprises at least two series connected DC link capacitors.

[0011] A DC bus or buses of the converter, preferably, include one or several DC link capacitors. There may be one DC link capacitor per DC bus, or several DC link capacitors connected in series and/or parallel per DC bus. The amount and connections between the DC link capacitors depend on the desired charging capacity and/or voltage level.

[0012] In various preferable embodiments, the pre-charging arrangement is configured to open the second circuit breaker after the pre-charging, and, subsequently, close the first circuit breaker for initiating a normal operation of the poly-phase power converter.

[0013] In various embodiments, the pre-charging arrangement, specifically the secondary side of the transformer thereof, may be connected between two star-points on the input side of the poly-phase power converter, namely the star-point of the filter capacitors of the input filter, and the star-point of the phase legs or the neutral point of the power converter.

[0014] The AC power supply may be a single-phase electric grid. For example, the primary side of the transformer may be connected between the line and the zero of the single-phase electric grid.

[0015] Alternatively or in addition, the AC power supply may comprise a pulse-width modulated (PWM) power supply connected via the transformer, preferably connected at the primary side thereof. Furthermore, the pulse-width modulated power supply may be configured to ramp-up a pre-charging voltage in accordance with a pre-defined slope. For example, a PWM regulated power-electronics converter may be used to act as an AC supply enabling controlled ramp-up/ramp-down of the AC voltage magnitude and/or frequency.

[0016] The primary side of the transformer may be connected to the AC power supply, and the secondary side of the transformer between the first common point and the second common point.

[0017] The poly-phase power converter as it is referred herein preferably comprises a plurality of phase legs, each one of the phase legs comprising a first terminal and a second terminal. The first terminals may be connected to respective phases of the poly-phase AC power supply and the second terminals are connected to the second common point. In case of a poly-phase cascaded H-bridge converter, there is a plurality of H-bridges connected in series between the first and second terminals in each phase leg.

[0018] The input filter preferably comprises at least a plurality of filter capacitors comprising first and second terminals, wherein the first terminals of the filter capacitors are connected to respective phases of the poly-phase AC power supply and the second terminals of the filter capacitors are connected to the first common point.

[0019] In some embodiments, the input filter is an LCL filter comprising the first common point at the second terminals of the filter capacitors.

[0020] Regarding the transformer, a turns ratio thereof, being a ratio of primary side turns to secondary side turns, is less than one. It may be even less than 0.8, 0.5, or 0.1, or even 0.01.

[0021] The transformer may be adapted to exhibit, during use thereof, a low voltage at the primary side. Said low voltage may be in the range from 100 V AC to 600 V AC, or even up to 1000 V AC, for instance, between the line and the neutral. Furthermore, the transformer may be adapted to exhibit, during use thereof, a medium or high voltage at the secondary side. Said medium voltage may be in the range from 1000 V AC or 1500 V AC up to 10000 or even 20000 V AC, for instance, between the line and the neutral. Said high voltage may be in the range from 10000 V AC upwards, for instance, between the line and the neutral.

[0022] In some embodiments, the transformer may be a double-winding or a multi-winding (more

than two) transformer to combine power flow among different phases in poly-phase applications.

[0023] The second circuit breaker may, preferably, be arranged on the primary side of the transformer. Thus, there is no need to have a medium or high voltage feed and contactor for the pre-charging arrangement, if the transformer is as described above, that is, having lower voltage on the primary than on the secondary side.

[0024] The first circuit breaker may be arranged between the poly-phase AC power supply and the input filter.

[0025] A common mode filter comprising a plurality of filter inductors may be connected between the input filter and the poly-phase H-bridge converter. The common mode filter may be of the type L, LC, LCL, or LLCL, for instance, where L refers to inductors and C to capacitors of the filter, as is understood by a skilled person.

[0026] According to a second aspect, a power converter system is provided. The power converter system comprises a poly-phase power converter, optionally a (cascaded) H-bridge converter, for being supplied by a poly-phase AC power supply, an input filter for the poly-phase power converter, the input filter comprising a first common point, and a first circuit breaker for connecting and disconnecting the poly-phase power converter and the input filter from the poly-phase AC power supply. The power converter system also comprises the pre-charging arrangement in accordance with the first aspect and/or any embodiment thereof as described in this document.

[0027] According to a third aspect, a method for pre-charging a poly-phase power converter is provided. The poly-phase power converter, optionally a (cascaded) H-bridge converter, is arranged to be supplied by a poly-phase AC power supply via an input filter, the input filter comprising a first common point, wherein the poly-phase power converter and the input filter are arranged to be connected and disconnected by a first circuit breaker. The method comprises pre-charging DC buses, preferably including one or several DC link capacitors therein, of the poly-phase power converter by electrical energy from an AC power supply via a transformer, having a primary side and a secondary side, connected between the first common point of the input filter and a second common point of the poly-phase power converter. The DC buses may, thus, comprise one or several DC link capacitors connected in series and/or parallel with each other.

[0028] In preferable embodiments, the method comprises closing a second circuit breaker to connect an AC power supply via a transformer, having a primary side and a secondary side, between the first common point of the input filter and a second common point of the poly-phase power converter for pre-charging DC link capacitors of the poly-phase power converter, and opening the second circuit breaker after the pre-charging.

[0029] The method may also comprise closing the first circuit breaker after the opening to initiate a normal operation of the poly-phase power converter.

[0030] The method may also comprise opening the second circuit breaker and then operating, such as pulsing or modulating, the power converter such that the filter input voltage gets synchronized with the grid. When the filter input voltage matches the grid in amplitude, frequency, and phase, that is it has been synchronized, the first circuit breaker may be closed.

[0031] The present invention provides a pre-charging arrangement, a power converter system, and a method for pre-charging a poly-phase power converter poly-phase power converter. The present invention provides advantages over known solutions in that it allows the use of a single pre-charge unit for the many DC buses in the poly-phase power converter resulting in substantial reduction in cost, size, and complexity of the system.

[0032] Various other advantages will become clear to a skilled person based on the following detailed description.

[0033] The expression “a number of” may herein refer to any positive integer starting from one (1).

[0034] The expression “a plurality of” may refer to any positive integer starting from two (2), respectively.

[0035] The terms “first” and “second” are herein used to distinguish one element from another

element, and not to specially prioritize or order them, if not otherwise explicitly stated.

[0036] The exemplary embodiments of the present invention presented herein are not to be interpreted to pose limitations to the applicability of the appended claims. The verb “to comprise” is used herein as an open limitation that does not exclude the existence of also unrecited features. The features recited in the appended patent claims are mutually freely combinable unless otherwise explicitly stated.

[0037] The novel features which are considered as characteristic of the present invention are set forth in particular in the appended claims. The present invention itself, however, both as to its construction and its method of operation, together with additional objectives and advantages thereof, will be best understood from the following description of specific embodiments when read in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0038] Some embodiments of the invention are illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings.

[0039] FIG. 1 illustrates schematically a pre-charging arrangement in connection with a poly-phase power converter.

[0040] FIG. 2 illustrates schematically a pre-charging arrangement in connection with a poly-phase power converter.

[0041] FIGS. 3A and 3B illustrates schematically H-bridge circuits.

[0042] FIG. 4 illustrates schematically a pre-charging arrangement in connection with a poly-phase power converter.

[0043] FIGS. 5A and 5B illustrate schematically AC power supplies of pre-charging arrangements.

[0044] FIGS. 6A and 6B shows graphs of pre-charging voltages and currents.

[0045] FIG. 7 shows a flow diagram of a method.

[0046] FIG. 8 illustrates schematically a pre-charging arrangement in connection with a poly-phase power converter.

DETAILED DESCRIPTION

[0047] FIG. 1 illustrates schematically a pre-charging arrangement **10** for a poly-phase, optionally (cascaded) H-bridge, power converter **20**, such a three-phase cascaded H-bridge converter, arranged for being supplied by a poly-phase AC power supply **40** via an input filter **30**.

Furthermore, the poly-phase, optionally (cascaded) H-bridge, converter **20** is arranged to be connected to and disconnected from the poly-phase AC power supply **40** by a first circuit breaker **35**. The poly-phase power converter **20** may comprise one or a plurality of H-bridge circuits **21A-23N** or other switching circuits in its plurality of phase legs **21-23**.

[0048] The pre-charging arrangement **10** comprises an AC power supply **12** connected via a transformer **14**, having a primary side **14A** and a secondary side **14B**, between a first common point **39** of the input filter **30** and a second common point **29** of the poly-phase, optionally (cascaded) H-bridge, converter **20** for pre-charging DC link capacitors **25** of the poly-phase power converter **20**. The pre-charging arrangement **10** also comprises a second circuit breaker **15** for connecting and disconnecting the pre-charging arrangement **10**.

[0049] Not depicted in the illustrations, the electronic circuit may incorporate a fuse connected in series with each AC power supply/poly-phase. This fuse is designed to break the power supply if the current surpasses a threshold defined by the fuse and its responsiveness. The fuse can be triggered to break in response to a fault condition in the power converter or any of the H-bridges, if any.

[0050] The pre-charging arrangement **10** and/or the poly-phase power converter **20** may comprise a

control unit **1000**, such as comprising a processing unit and a memory, as well as inputs and outputs for receiving data, such as measurement data, and for outputting command signals, such as to open and/or close circuit breakers **15**, **35**. The control unit **1000** may only control the pre-charging or, optionally, also the poly-phase power converter **20**. Alternatively, there may be dedicated control units for both of them. In some cases, the control unit **1000** is comprised in the poly-phase power converter **20** and is arranged to control the pre-charging also.

[0051] In addition, the pre-charging arrangement **10** may be configured to open the second circuit breaker **15** after the pre-charging, and, subsequently, close the first circuit breaker **35** for initiating a normal operation of the poly-phase power converter **20**.

[0052] In addition, preferably, the pre-charging arrangement **10** may be configured to close the second circuit breaker **15** to initiate the pre-charging. In such cases, the pre-charging arrangement **10** may be configured to keep the first circuit breaker **35** open, or even actively prevent it from being closed during the pre-charging.

[0053] The AC power supply **40** may be a single-phase electric grid as illustrated in FIG. 5A. The primary side **14A** of the transformer **14** may be connected between the line and the zero of the single-phase electric grid.

[0054] In addition or alternatively, the AC power supply **40** may comprise or be a pulse-width modulated power supply **16** connected to the transformer **14**. This is illustrated in FIG. 5B. The pulse-width modulated power supply **16** may be configured to ramp-up pre-charging voltage in accordance with a pre-defined slope.

[0055] Furthermore, the primary side **14A** of the transformer **14** may be connected to the AC power supply **12**, and the secondary side **14B** of the transformer **14** between the first common point **39** and the second common point **29**.

[0056] The poly-phase power converter **20** may comprise a plurality of phase legs **21-23**, each one of the phase legs **21-23** comprising a first terminal **T1_21-T1_23** and a second terminal **T2_21-T2_23**, and, optionally, one or a plurality of H-bridges **21A-21N**, **22A-22N**, **23A-23N** or other switching circuits/bridges connected in series between the first **T1_21-T1_23** and second terminals **T2_21-T2_23**, wherein the first terminals **T1_21-T1_23** are connected to respective phases **41-43** of the poly-phase AC power supply **40** and the second terminals **T2_21-T2_23** are connected to the second common point **29**. In case of having a plurality of H-bridges **21A-21N**, **22A-22N**, **23A-23N** connected in series between the first **T1_21-T1_23** and second terminals **T2_21-T2_23**, the converter **20** may be called a poly-phase cascaded H-bridge converter **20**.

[0057] The input filter **30** may comprise at least a plurality of filter capacitors **32** comprising first **32A** and second terminals **32B**, wherein the first terminals **32A** of the filter capacitors **32** are connected to respective phases **41-43** of the poly-phase AC power supply **40** and the second terminals **32B** of the filter capacitors **32** are connected to the first common point **39** and/or to form the first common point **39**.

[0058] In some embodiments, the input filter **30** may be an LCL filter comprising the first common point **39** at the second terminals **32B** of the filter capacitors **32**.

[0059] A turns ratio of the transformer **14**, being a ratio of primary side turns to secondary side turns, may be less than one. It may be even less than 0.8, 0.5, or 0.1, or even 0.01.

[0060] The transformer may be adapted to exhibit, during use thereof, a low voltage at the primary side. Said low voltage may be in the range from 100 V AC to 600 V AC, or even up to 1000 V AC, for instance, between the line and the neutral. Furthermore, the transformer may be adapted to exhibit, during use thereof, a medium or high voltage at the secondary side. Said medium voltage may be in the range from 1000 V AC or 1500 V AC up to 10000 or even 20000 V AC, for instance, between the line and the neutral. Said high voltage may be in the range from 10000 V AC upwards, for instance, between the line and the neutral.

[0061] The second circuit breaker **15** may be arranged on the primary side **14A** of the transformer **14** as shown in FIG. 1. Thus, the second circuit breaker **15** may be a low voltage circuit breaker.

There is thus no need to add medium or high voltage circuit breakers to the power converter system **100** in order to perform the pre-charging.

[0062] Regarding the first circuit breaker **35**, it may, preferably, be arranged between the poly-phase AC power supply **40** and the input filter **30**.

[0063] FIG. **2** illustrates schematically a pre-charging arrangement **10** in connection with a poly-phase power converter **20**. The pre-charging arrangement **10** may be otherwise similar to one shown in FIG. **1**, however, the second circuit breaker **15** is arranged to the secondary side **14B** of the transformer **14**.

[0064] FIGS. **3A** and **3B** illustrates schematically H-bridges or H-bridge circuits **21A-23N**. FIG. **3A** illustrates a two-level H-bridge. FIG. **3B** illustrates a three-level H-bridge. The present invention is not limited to two- and three-level H-bridges but may be implemented with H-bridges having any level, such as four-level, five-level, six-level etc. The H-bridges may comprise NP or NPC (Neutral Point Clamped) topologies.

[0065] FIG. **4** illustrates schematically a pre-charging arrangement **10** in connection with a poly-phase power converter **20**. The pre-charging arrangement **10** may be similar to one shown in FIG. **1** or FIG. **2** but there is a common mode filter **50** comprising a plurality of filter inductors, such as coupled inductors, connected between the input filter **30** and the poly-phase power converter **20**.

[0066] In various embodiments, such as illustrated in FIGS. **1**, **2**, and **4**, the first common point **39** may be connected to the ground potential. The connection may be a low resistance or solid grounding connection, that is being more or less directly connected to grounding wire or the like. On the other hand, FIG. **4** illustrates, which may apply to other embodiments as well, a common mode capacitor **52** via which the first common point **39** may be grounded.

[0067] In view of the above, there is also disclosed herein a power converter system **100** comprising the poly-phase, optionally (cascaded) H-bridge, power converter **20** for being supplied by a poly-phase AC power supply **40**, the input filter **30** for the poly-phase power converter **20**, the input filter **30** comprising a first common point **39**, the first circuit breaker **35** for connecting and disconnecting the poly-phase power converter **20** and the input filter **30** from the poly-phase AC power supply **40**, and the pre-charging arrangement **10** as described herein above. The power converter system **100** may comprise its own control unit **1000**, or the control unit **1000** may be comprised in the converter **20** and/or the pre-charging arrangement **10**.

[0068] FIGS. **6A** and **6B** shows graphs of pre-charging voltages **101A**, **102A** and currents **101B**, **102B**. In embodiments with a single-phase electric grid as the AC power supply **12**, the amplitude of the pre-charging voltage may be as shown by **101A**. The pre-charging voltage **101A** represents the voltage of the DC link capacitors **25** of the power converter **20** or of an H-bridge **21A-23N**, or the series connection thereof. The pre-charging current in such case is shown by **101B**. The pre-charging current **101B** includes a higher value portion before time instance **103**. After that the current stabilizes to a lower value.

[0069] In another examples, wherein the pre-charging arrangement **10** comprise a controllable AC power supply **12**, such as a pulse-width modulated power supply **16**, e.g. including a PWM regulated power converter, the pre-charging current **102B** may be controlled so as not to become too high. As can be seen, marked by **102A**, the pre-charging voltage then increases slower. The pulse-width modulated power supply **16** may thus be configured to ramp-up pre-charging voltage in accordance with a pre-defined slope.

[0070] FIG. **7** shows a flow diagram of a method. The method is preferably for pre-charging a poly-phase, optionally (cascaded) H-bridge, power converter **20** arranged to be supplied by a poly-phase AC power supply **40** via an input filter **30**, the input filter **30** comprising a first common point **39**, and wherein the poly-phase power converter **20** and the input filter **30** are arranged to be connected and disconnected by the first circuit breaker **35**.

[0071] Item or step **700** refers to an optional start-up phase of the method. Suitable equipment and components are obtained, and systems assembled and configured for operation.

[0072] The method steps may, in accordance with a non-limiting example, be performed by the control unit **1000** as described hereinabove, for instance.

[0073] Item or step **710** refers to pre-charging DC link capacitors **25** of the poly-phase power converter **20** by electrical energy from an AC power supply **12** via a transformer **14**, having a primary side **14A** and a secondary side **14B**, connected between the first common point **39** of the input filter **30** and a second common point **29** of the poly-phase power converter **20**.

[0074] In various embodiments, the pre-charging may comprise opening the second circuit breaker **15** after the pre-charging, and the method further comprises closing the first circuit breaker **35** after said opening to initiate a normal operation of the poly-phase power converter **20**.

[0075] Furthermore, optionally, the pre-charging may comprise closing the second circuit breaker **15** to connect the AC power supply **12** via the transformer **14** between the first common point **39** and the second common point **29** to start pre-charging of the DC link capacitors **25** of the poly-phase converter **20**.

[0076] In accordance with an embodiment, the method comprises, in stated the order: [0077] 1) closing the second circuit breaker or contactor **15**, [0078] 2) charging the DC link capacitors **25** to a desired voltage level, [0079] 3) opening the second circuit breaker or contactor **15**, and [0080] 4) closing the first circuit breaker (which has been open up to this step).

[0081] There may be an optional step **3b**) after step **3**) and before step **4**), during which the input filter is synchronized with respect to the poly-phase AC power supply **40** as described hereinbefore.

[0082] After step **4**) above, the normal operation may be started constituting to the end of the pre-charging.

[0083] Method execution may be stopped at item or step **799**.

[0084] Term “normal operation of the poly-phase power converter” refers to operation of the poly-phase power converter **20** after the pre-charging during which the converter is, for example, utilized and operated to produce the desired voltage(s) and current(s) at its output. On the other hand, the normal operation may also include such that the poly-phase power converter **20** is not necessarily producing any voltage(s) and current(s) but it may only be configured to be ready to do so after the pre-charging, that is, the converter is in a ready state having the DC link capacitors **25** pre-charged suitably and/or to a pre-defined value, and having the first circuit breaker **35** closed.

[0085] FIG. **8** illustrates schematically a pre-charging arrangement **100** in connection with a poly-phase power converter **20**. In FIG. **8**, the power converter **20** may be an NPC power converter. As can be seen, the second common point **29** is the neutral point or the DC link midpoint of the power converter **20**. The NPC power converter comprises a plurality of phase legs **21-23**, however, they are not illustrated in detail in FIG. **8** for simplicity. The skilled person realizes that the power converter **20** as illustrated in FIG. **8** may comprise three phase legs **21-23** comprising controllable switching devices which are controlled, such as by utilizing a PWM technique, to produce desired DC output across the DC link capacitors **25**. Thus, in some embodiments, it is not necessary to have separate switching circuits or bridges (such as shown in FIGS. **1**, **2**, and **4**) for each phase.

[0086] As understood by the skilled person, the power converter **20** and/or the converter system **100** may comprise various sensors for measuring voltages and currents in the converter **20** and/or the system **100**. For example, input and output voltages and currents, and DC bus voltages etc. The measurements may be received at the control unit **1000**. Furthermore, the control unit **1000** may be adapted to control circuit breakers **15**, **35** and optionally other devices/components.

[0087] While the present disclosure has been illustrated and described with respect to a particular embodiment thereof, it should be appreciated by those of ordinary skill in the art that various modifications to this disclosure may be made without departing from the spirit and scope of the present disclosure.

Claims

1. A pre-charging arrangement for a poly-phase power converter arranged for being supplied by a poly-phase AC power supply via an input filter, wherein the poly-phase power converter is arranged to be connected to and disconnected from the poly-phase AC power supply by a first circuit breaker, wherein the pre-charging arrangement comprises: an AC power supply connected via a transformer, having a primary side and a secondary side, between a first common point of the input filter and a second common point of the poly-phase power converter for pre-charging DC link capacitors of the poly-phase power converter, and a second circuit breaker for connecting and disconnecting the pre-charging arrangement.
2. The pre-charging arrangement of claim 1, configured to: open the second circuit breaker after the pre-charging, and, subsequently, close the first circuit breaker for initiating a normal operation of the poly-phase power converter.
3. The pre-charging arrangement of claim 1, wherein the AC power supply is a single-phase electric grid.
4. The pre-charging arrangement of claim 1, wherein the AC power supply comprises a pulse-width modulated power supply connected to the transformer.
5. The pre-charging arrangement of claim 4, wherein the pulse-width modulated power supply is configured to ramp-up a pre-charging voltage in accordance with a pre-defined slope.
6. The pre-charging arrangement of claim 1, wherein the primary side of the transformer is connected to the AC power supply, and the secondary side of the transformer between the first common point and the second common point.
7. The pre-charging arrangement of claim 1, wherein the poly-phase power converter comprises a plurality of phase legs, each one of the phase legs comprising a first terminal (T1_21-T1_23) and a second terminal (T2_21-T2_23), and a plurality of H-bridges connected in series between the first (T1_21-T1_23) and second terminals (T2_21-T2_23), wherein the first terminals (T1_21-T1_23) are connected to respective phases of the poly-phase AC power supply and the second terminals (T2_21-T2_23) are connected to the second common point.
8. The pre-charging arrangement of claim 1, wherein the input filter comprises at least a plurality of filter capacitors comprising first (T1_21-T1_23) and second terminals (T2_21-T2_23), wherein the first terminals (T1_21-T1_23) of the filter capacitors are connected to respective phases of the poly-phase AC power supply and the second terminals (T2_21-T2_23) of the filter capacitors are connected to the first common point.
9. The pre-charging arrangement of claim 1, wherein a turns ratio of the transformer, being a ratio of primary side turns to secondary side turns, is less than one.
10. The pre-charging arrangement of claim 1, wherein the second circuit breaker is arranged on the primary side of the transformer.
11. The pre-charging arrangement of claim 1, wherein the first circuit breaker is arranged between the poly-phase AC power supply and the input filter.
12. The pre-charging arrangement of claim 1, wherein a common mode filter comprising a plurality of filter inductors is connected between the input filter and the poly-phase converter.
13. A power converter system, comprising: a poly-phase power converter, such as an optionally cascaded, H-bridge converter, for being supplied by a poly-phase AC power supply, an input filter for the poly-phase power converter, the input filter comprising a first common point, a first circuit breaker for connecting and disconnecting the poly-phase power converter and the input filter from the poly-phase AC power supply, and the pre-charging arrangement of claim 1.
14. A method for pre-charging a poly-phase power converter, such as an, optionally cascaded, H-bridge converter, arranged to be supplied by a poly-phase AC power supply via an input filter, the input filter comprising a first common point, wherein the poly-phase power converter and the input filter are arranged to be connected and disconnected by a first circuit breaker, the method comprising: pre-charging DC link capacitors of the poly-phase power converter by electrical

energy from an AC power supply via a transformer, having a primary side and a secondary side, connected between the first common point of the input filter and a second common point of the poly-phase power converter.

15. The method of claim 14, wherein the pre-charging comprises: closing a second circuit breaker to connect an AC power supply via a transformer, having a primary side and a secondary side, between the first common point of the input filter and a second common point of the poly-phase power converter for pre-charging DC link capacitors of the poly-phase power converter, and opening the second circuit breaker after the pre-charging, wherein, the method comprises closing the first circuit breaker after the opening to initiate a normal operation of the poly-phase power converter.
